

Java

Widely used in companies and placement tests · Equigrow short notes

Hello & basics

```
public class Main {  
    public static void main(String[] args) {  
        int age = 18; double cgpa = 8.5; String name = "Asha";  
        System.out.println(name + " " + age);  
    }  
}
```

- File name must match the public class: Main.java → javac Main.java then java Main.
- Statically typed like C; memory is managed for you (garbage collection).

Input, loops, methods

```
import java.util.Scanner;  
Scanner sc = new Scanner(System.in);  
int n = sc.nextInt();  
for (int i = 0; i < n; i++) { ... }  
static int square(int x) { return x * x; }
```

OOP

```
class Student {  
    private String name; // encapsulation  
    Student(String name) { this.name = name; } // constructor  
    String getName() { return name; }  
}  
class Topper extends Student { ... } // inheritance
```

- Interfaces define what a class can do; abstract classes share partial code.
- Overloading = same name, different parameters. Overriding = subclass changes a method.

Collections

Type	Use
ArrayList<Integer>	growing list
HashMap<String,Integer>	key → value, fast lookup
HashSet<String>	unique items
ArrayDeque	stack / queue